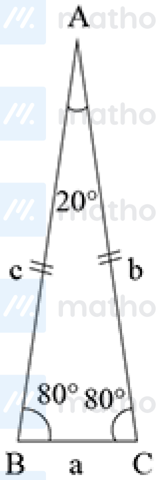
Q1 (3) - Single Correct

We have, in $\Delta ABC, b=c$, then
$\angle B=\angle C$
$\therefore \angle B=\angle C=\frac{160^{\circ}}{2}=80^{\circ}$
$(\because \angle A=20^{\circ}, \text{ given })$



Now, using sine rule, we get

$\frac{a}{\sin 20^{\circ}}=\frac{b}{\sin 80^{\circ}}=\frac{c}{\sin 80^{\circ}}$

If $\frac{a}{\sin 20^{\circ}}=\frac{b}{\sin 80^{\circ}}=\frac{b}{\sin \left(90^{\circ}-10^{\circ}\right)}$

$\Rightarrow \quad \frac{a}{\sin 20^{\circ}}=\frac{b}{\cos 10^{\circ}}$

$\Rightarrow \quad a=\frac{b \sin 20^{\circ}}{\cos 10^{\circ}}=\frac{2 b \cdot \sin 10^{\circ} \cos 10^{\circ}}{\cos 10^{\circ}}$

$\Rightarrow \quad a=2 b \sin 10^{\circ} \quad$ (i)

Thus, $\quad a^3+b^3=\left(2 b \sin 10^{\circ}\right)^3+b^3$

$=8 b^3 \sin ^3 10^{\circ}+b^3$

$=2 b^3\left(4 \sin ^3 10^{\circ}\right)+b^3$

$=2 b^3\left(3 \sin 10^{\circ}-\sin 30^{\circ}\right)+b^3 \quad\left(\because \sin 3 A=3 \sin A-4 \sin ^3 A\right)$

$=3 b^2\left(2 b \sin 10^{\circ}\right)-b^3+b^3$

$=3 b^2 \cdot a \quad$ [from Eq. (i)]

$=3 c^2 a \quad(\because b=c)$

Q2 (2) - Single Correct

$\because a, b, c$ are in AP, then

$2b = a + c$

$2 \times 5 = 3 + c \quad (\because a = 3, b = 5)$

$c = 10 - 3 = 7$

$$\therefore \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$= \frac{3^2 + 5^2 - 7^2}{2 \times 3 \times 5}$$

$$= \frac{-1}{2} = \cos 120^\circ$$

$\therefore C = 120^\circ$, then $A, B < 60^\circ$, then $\tan A, \tan B > 0$

$\Rightarrow \tan A + \tan B > 0$ ...(i)

We know that,

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$\Rightarrow \tan A + \tan B + \tan 120^\circ = \tan A \tan B \tan 120^\circ$$

$$\Rightarrow \tan A + \tan B - \sqrt{3} = -\sqrt{3} \tan A \tan B$$

$\Rightarrow \quad \tan A + \tan B = \sqrt{3}(1 - \tan A \tan B)$ ...(ii)

$\therefore \sqrt{3}(1 - \tan A \tan B) > 0 \quad$ [from Eq.(i)]

$\tan A \tan B < 1$ ...(iii)

We know that,

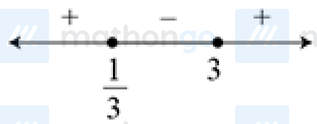
$AM \geq GM$

$\frac{\tan A + \tan B}{2} \geq \sqrt{\tan A \tan B}$

$\sqrt{3}(1 - \tan A \tan B) \geq 2\sqrt{\tan A \tan B} \quad$ [from Eq.(ii)]

Let $\tan A \tan B = y$

$\sqrt{3}(1 - y) \geq 2\sqrt{y}$



$3\left(1 + y^2 - 2y\right) \geq 4y$

$3y^2 - 10y + 3 \geq 0$

$(3y - 1)(y - 3) \geq 0$

$y \leq \frac{1}{3}$ and $y \geq 3$

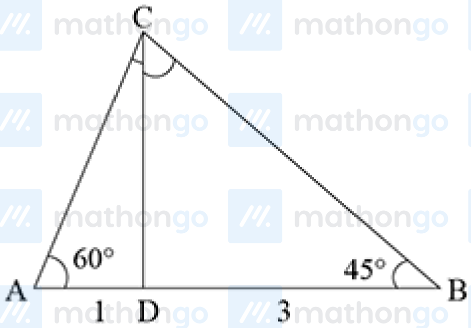
$\tan A \tan B \leq \frac{1}{3}, \tan A \tan B \geq 3$ (neglecting)

$\therefore$ Maximum value of $\tan A \tan B = \frac{1}{3}$

Q3 (3) - Single Correct

$\because D$ divides AB in 1 : 3.

Let $AB = a$, then $AD = \frac{a}{4}, BD = \frac{3}{4}a$



Using sine rule, in ΔACD and ΔBCD,

$\frac{CD}{\sin 60^{\circ}} = \frac{AD}{\sin(\angle ACD)}$

and $\frac{CD}{\sin 45^{\circ}} = \frac{BD}{\sin(\angle BCD)}$

$CD = \frac{\sqrt{3}}{2} \cdot \frac{a}{4\sin(\angle ACD)}$ ...(i)

and $CD = \frac{1}{\sqrt{2}} \cdot \frac{3a}{4\sin(\angle BCD)}$ ...(ii)

From Eqs. (i) and (ii), we get

$\frac{1}{\sqrt{2}} \cdot \frac{3a}{4\sin(\angle BCD)} = \frac{\sqrt{3}a}{2\times 4\sin(\angle ACD)}$

$\frac{\sin(\angle ACD)}{\sin(\angle BCD)} = \frac{1}{\sqrt{6}}$

$\therefore \quad \frac{\sqrt{6}\sin(\angle ACD)}{\sin(\angle BCD)} = 1$

Q4 (1) - Single Correct

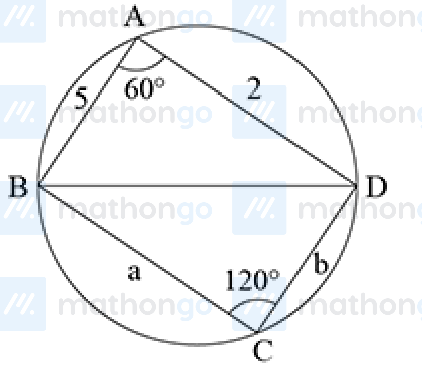
Let $BC = a$ and $CD = b$

$\operatorname{ar}(ABCD) = \operatorname{ar}(\Delta ABD) + \operatorname{ar}(\Delta BCD)$

$4\sqrt{3} = \frac{1}{2} \times 2 \times 5 \cdot \sin 60^{\circ} + \frac{1}{2}ab\sin 120^{\circ}$

$4\sqrt{3} = \frac{5\sqrt{3}}{2} + \frac{ab\sqrt{3}}{4}$

$ab = 6$ ...(i)



and we know that,

$$\cos A=\frac{2^2+5^2-a^2-b^2}{2(10+ab)}$$

$$\Rightarrow \quad \cos 60^{\circ}=\frac{29-a^2-b^2}{2(10+ab)}$$

$\frac{1}{2}=\frac{29-a^2-b^2}{2(10+6)}$ [from Eq.(i)]

$$\Rightarrow \quad a^2+b^2=13 \quad \ldots\text{(ii)}$$

$$\because \quad (a+b)^2=a^2+b^2+2ab$$
$$=13+2\times 6=25$$

Now, $\quad a+b=5$ and $ab=6$

On solving, we get

$a=3, b=2$

or $a=2, b=3$

Q5 (3) - Single Correct

We know that,

$(a-b)^2+(b-c)^2+(c-a)^2\geq 0$

$2\left(a^2+b^2+c^2\right)\geq 2(ab+bc+ca)$

$3\left(a^2+b^2+c^2\right)\geq a^2+b^2+c^2+2ab+2bc+2ca$

$3\left(a^2+b^2+c^2\right)\geq(a+b+c)^2>d^2$

$\frac{a^2+b^2+c^2}{d^2}>\frac{1}{3}$

Q6 (1) - Single Correct

We have

$p=\sin A \sin B \sin C$

$q=\cos A \cos B \cos C$

$\therefore \quad \tan A \tan B \tan C=\frac{p}{q}$ ...(i)

and $\tan A \tan B+\tan B \tan C+\tan C \tan A$
$=\frac{\sin A \sin B \cos C+\sin B \sin C \cos A+\sin A \cos B \sin C}{\cos A \cos B \cos C}$
$=\frac{\sin B(\sin A \cos C+\cos A \sin C)+\sin A \cos B \sin C}{\cos A \cos B \cos C}$
$=\frac{\sin B \sin (A+C)+\sin A \cos B \sin C}{\cos A \cos B \cos C}$
$=\frac{\sin ^2 B+\sin A \cos B \sin C}{\cos A \cos B \cos C}$
$=\frac{1-\cos ^2 B+\sin A \cos B \sin C}{\cos A \cos B \cos C}$
$=\frac{1-\cos B(\cos B-\sin A \sin C)}{\cos A \cos B \cos C}$
$=\frac{1+\cos B[\cos (A+C)+\sin A \sin C]}{\cos A \cos B \cos C}$
$=\frac{1+\cos B[\cos A \cos C-\sin A \sin C+\sin A \sin C]}{\cos A \cos B \cos C}$
$=\frac{1+\cos B \cos A \cos C}{\cos A \cos B \cos C}$
$=\frac{1+q}{q}$

We know that, in ΔABC

$\tan A+\tan B+\tan C=\tan A \tan B \tan C=\frac{p}{q}$

Let $\tan A=\alpha, \tan B=\beta, \tan C=\gamma$, then the equation whose roots are α, β, γ is

$x^3-(\alpha+\beta+\gamma) x^2+(\alpha \beta+\beta \gamma+\gamma \alpha) x-\alpha \beta \gamma=0$

$x^3-\frac{p}{q} x^2+\left(\frac{1+q}{q}\right) x-\frac{p}{q}=0$

$\therefore \quad q x^3-p x^2+(1+q) x-p=0$

Q7 (1) - Single Correct

We have, $F(x+y)=F(x) \cdot F(y), \forall x, y$ and $F(1)=2$

$F(2)=F(1+1)=F(1) \cdot F(1)=2 \times 2=4$
$F(3)=F(1+2)=F(1) \cdot F(2)=2 \times 4=8$
$\therefore a=F(3)=8, b=F(1)+F(3)=2+8=10$
$c=F(2)+F(3)=4+8=12$

Using cosine rule,

$$\cos A=\frac{b^2+c^2-a^2}{2bc}=\frac{(10)^2+(12)^2-(8)^2}{2\times 10\times 12}$$
$$=\frac{100+144-64}{240}=\frac{180}{240}$$
$$\cos A=\frac{3}{4}$$
$$\therefore\quad \cos 2A=2\cos^2 A-1=2\left(\frac{3}{4}\right)^2-1$$
$$\cos 2A=\frac{1}{8}=\cos C$$
$$\left(\because \cos C=\frac{a^2+b^2-c^2}{2ab}=\frac{64+100-144}{2\times 8\times 10}=\frac{1}{8}\right)$$
$$\therefore\quad 2A=C$$

Q8 (1) - Single Correct

$\because r_1=\frac{\Delta}{s-a}, r_2=\frac{\Delta}{s-b}, r_3=\frac{\Delta}{s-c}$

Now, $\lambda\left(s\Sigma\frac{1}{a}-3\right)=\frac{bc}{r_1}+\frac{ca}{r_2}+\frac{ab}{r_3}$

$=\frac{bc(s-a)}{\Delta}+\frac{ca(s-b)}{\Delta}+\frac{ab(s-c)}{\Delta}$

$=\frac{abc}{\Delta}\left(\frac{s-a}{a}+\frac{s-b}{b}+\frac{s-c}{c}\right)$

$=\frac{abc}{\Delta}\left(\frac{s}{a}+\frac{s}{b}+\frac{s}{c}-3\right)$

$=\frac{abc}{\Delta}\left[s\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)-3\right]$

$$\lambda\left(s\Sigma\frac{1}{a}-3\right)=\frac{abc}{\Delta}\left(s\sum\frac{1}{a}-3\right)$$
$$\therefore\quad \lambda=\frac{abc}{\Delta}$$

Q9 (2) - Single Correct

$\because \cos A+\cos B+\cos C=\frac{7}{4}$

$1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}=\frac{7}{4}$

(using $\cos A+\cos B+\cos C=1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$)

$\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}=\frac{3}{16}$ ...(i)

We know that

$r=4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$=4R\times\frac{3}{16}=\frac{3R}{4}$ [from Eq.(i)]

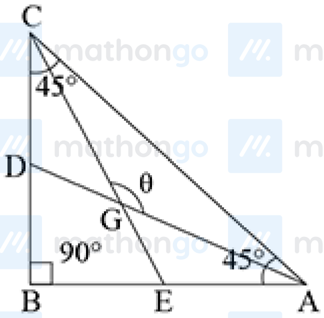
$\Rightarrow \quad \frac{R}{r}=\frac{4}{3}$

$\therefore \quad \frac{27 R}{r}=27 \cdot \frac{4}{3}=36$

Q10 (3) - Single Correct

Let $AB=BC=k$
$\Rightarrow AC=\sqrt{2} k$

and let θ be the obtuse angle formed by medians



$\because AD$ and CE are medians, then G is the centroid and $BE=\frac{k}{2}$.

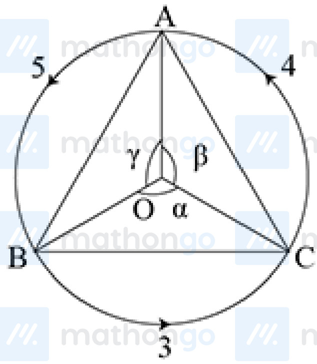
$\because GA=GC=\frac{2}{3} CE=\frac{2}{3} \sqrt{(BC)^2+(BE)^2}$
(using Pythagoras theorem in ΔBCE)
$=\frac{2}{3} \sqrt{k^2+\frac{k^2}{4}}=\frac{2}{3} \cdot \frac{\sqrt{5}}{2} k=\frac{\sqrt{5}}{3} k$

Now, applying cosine rule, in ΔAGC,

$$\begin{aligned}
\cos \theta & =\frac{(AG)^2+(GC)^2-(AC)^2}{2(AG)(GC)} \\
& =\frac{\frac{5}{9} k^2+\frac{5}{9} k^2-2 k^2}{2 \cdot \frac{\sqrt{5}}{3} k \cdot \frac{\sqrt{5}}{3} k} \\
& =\frac{-8 k^2}{10 k^2}=\frac{-4}{5}
\end{aligned}$$

Q11 (1) - Single Correct

$\because$ Arc$(BC) = 3$, Arc$(CA) = 4$ and Arc$(AB) = 5$



Let r be the radius of circle.

We know that, Arc $=$ Radius $\times$ Angle

Then, $3 = r\alpha, 4 = r\beta, 5 = r\gamma$

$\Rightarrow \quad \alpha = \frac{3}{r}, \beta = \frac{4}{r}, \gamma = \frac{5}{r}$

and $\quad 2\pi r = 3 + 4 + 5$

$\Rightarrow \quad r = \frac{6}{\pi}$

$\because \quad \text{ar}(\Delta ABC) = \text{ar}(\Delta OBC) + \text{ar}(\Delta OAC) + \text{ar}(\Delta OAB)$

$= \frac{1}{2}r^2 \sin\alpha + \frac{1}{2}r^2 \sin\beta + \frac{1}{2}r^2 \sin\gamma$

$= \frac{r^2}{2}\left(\sin\frac{3}{r} + \sin\frac{4}{r} + \sin\frac{5}{r}\right)$

$= \frac{1}{2} \cdot \frac{36}{\pi^2}\left(\sin\frac{\pi}{2} + \sin\frac{2\pi}{3} + \sin\frac{5\pi}{6}\right)$

$= \frac{18}{\pi^2}\left(1 + \frac{\sqrt{3}}{2} + \frac{1}{2}\right)$

$= \frac{9\sqrt{3}(1+\sqrt{3})}{\pi^2}$ sq units

Q12 (1,2,3) - Multiple Correct

We have,

$\frac{a\cos A + b\cos B + c\cos C}{a\sin B + b\sin C + c\sin A} = \frac{a+b+c}{9R}$

$\frac{2R\sin A\cos A + 2R\sin B\cos B + 2R\sin C\cos C}{a\cdot\frac{b}{2R} + b\cdot\frac{c}{2R} + c\cdot\frac{a}{2R}} = \frac{a+b+c}{9R}$

(using $\frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = R$)

$\frac{2R^2(\sin 2A + \sin 2B + \sin 2C)}{ab+bc+ca} = \frac{a+b+c}{9R}$

$\frac{2R^2(4\sin A\sin B\sin C)}{ab+bc+ca} = \frac{a+b+c}{9R}$

$\frac{8R^3\cdot\frac{a}{2R}\frac{b}{2R}\cdot\frac{c}{2R}}{ab+bc+ca} = \frac{a+b+c}{9R}$

$\Rightarrow 9abc=(a+b+c)(ab+bc+ca)$

$\Rightarrow a(b-c)^2+b(c-a)^2+c(a-b)^2=0$

which is possible only when $a-b=0$,

$b-c=0$,

and $c-a=0$

$\therefore a=b=c$

$\therefore \Delta ABC$ is an equilateral triangle,

i.e. $\angle A=\angle B=\angle C=60^{\circ}$

Q13 (1,2,4) - Multiple Correct

$\because r=\frac{\Delta}{s}, r_1=\frac{\Delta}{s-a}$

Now, $r_1 \cdot r=\frac{\Delta^2}{s(s-a)}=\frac{s(s-a)(s-b)(s-c)}{s(s-a)}$

$=(s-b)(s-c)=(s-b)^2$

$=\frac{(2s-2b)^2}{4}=\frac{(a+b+c-2b)^2}{4}$

$=\frac{a^4}{4}$ [option (d)]

$=\frac{4R^2\sin^2 A}{4}=R^2\sin^2 A$ [option (a)]

Also, if $\angle B=\theta$

$\Rightarrow \angle A=\pi-2\theta$

$\therefore r_1 \cdot r=R^2\sin^2(\pi-2\theta)$

$=R^2\sin^2 2\theta$

$=R^2\sin^2 2B$ [option (b)]

Q14 (1,3,4) - Multiple Correct

Given that, $s-a, s-b, s-c$ are in AP.

$\Rightarrow$ a, b, c are in AP. [option (c)]

$\Rightarrow \frac{\Delta}{s-a}, \frac{\Delta}{s-b}, \frac{\Delta}{s-c}$ are in HP.

$\Rightarrow r_1, r_2, r_3$ are in HP. [option (a)]

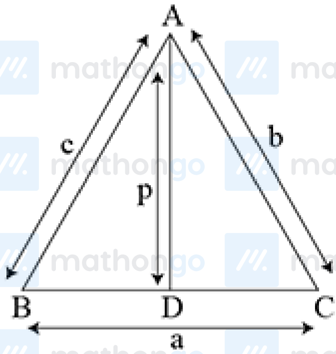
Also, $\cos A=\frac{b^2+c^2-a^2}{2bc}$

$=\frac{b^2+c^2-(2b-c)^2}{2bc}$

$=\frac{4c-3b}{2c}$ [option (d)]

Q15 (1,2,3) - Multiple Correct

Let AD be the median, then $BD = CD = \frac{a}{2}$



$\because AB^2 + AC^2 = 2\left(AD^2 + BD^2\right)$ (using median theorem)

$$c^2 + b^2 = 2\left(p^2 + \frac{a^2}{4}\right)$$

$\Rightarrow \quad b^2 + c^2 = 2p^2 + \frac{a^2}{2}$

$\Rightarrow \quad 4p^2 = 2b^2 + 2c^2 - a^2 \quad$ [option (a)]

$= b^2 + c^2 + \left(b^2 + c^2 - a^2\right)$

$\Rightarrow \quad 4p^2 = b^2 + c^2 + 2bc\cos A \quad$ [option (b)]

$= \left(b^2 + c^2 - a^2\right) + 2bc\cos A + a^2$

$= 2bc\cos A + 2bc\cos A + a^2$

$4p^2 = a^2 + 4bc\cos A \quad$ [option (c)]

Q16 (2,4) - Multiple Correct

Let a.b.c be the sides of ΔABC. then

$\cos A = \frac{b^2+c^2-a^2}{2bc}$ (using cosine rule)

$\Rightarrow \quad c^2 - 2b\cos A \cdot c + b^2 - a^2 = 0$ ...(i)

(which is a quadratic equation in C)

Let c, and c_2 be two roots of Eq. (i), then

$c_1 + c_2 = 2b\cos A \quad$ ...(ii)

$c_1 \cdot c_2 = b^2 - a^2 \quad$ ...(iii)

We have. $c_1^2 + c_1 \cdot c_2 + c_2^2 = a^2$

$c_1^2 + c_2^2 + c_1 \cdot c_2 = a^2$

$\Rightarrow (c_1 + c_2)^2 - c_1c_2 = a^2$

$\Rightarrow \quad 4b^2\cos^2 A - b^2 + a^2 = a^2 \quad$ [from Eqs. (ii) and (iii)]

$\Rightarrow \quad \cos^2 A=\frac{1}{4}$

$\Rightarrow \quad \cos A=\pm\frac{1}{2}$

$\therefore \quad \angle A=60^{\circ}$ and 120°

Q17 (2,3,4) - Multiple Correct

We have, $r_1=8, r_2=12, r_3=24$, then

$r_1 r_2+r_2 r_3+r_3 r_1=96+288+192=576$

We know that,

$a=\frac{r_1(r_2+r_3)}{\sqrt{r_1 r_2+r_2 r_3+r_3 r_1}}=\frac{8(12+24)}{\sqrt{576}}=\frac{8\times 36}{24}=12 \quad$ [option (b)]

$b=\frac{r_2(r_1+r_3)}{\sqrt{r_1 r_2+r_2 r_3+r_3 r_1}}=\frac{12(8+24)}{24}=\frac{12\times 32}{24}=16$ [option (c)]

$c=\frac{r_3(r_1+r_2)}{\sqrt{r_1 r_2+r_2 r_3+r_3 r_1}}=\frac{24(8+12)}{24}=\frac{24\times 20}{24}=20$ [option (d)]

Now, $\quad s=\frac{a+b+c}{2}=\frac{12+16+20}{2}=24$

We know that, $r_1 r_2 r_3=r s^2$

$8\times 12\times 24=r(24)^2$

$\therefore \quad r=\frac{8\times 12\times 24}{(24)^2}=4$ [option (a) is not correct]

Q18 (1,2,3,4) - Multiple Correct

Let r be the inradius and r_1, r_2, r_3 be the exradii, then

$A=\pi r^2, A_1=\pi r_1^2, A_2=\pi r_2^2, A_3=\pi r_3^2$

Now, option (a), LHS $=\sqrt{A_1}+\sqrt{A_2}+\sqrt{A_3}$

$=\sqrt{\pi} r_1+\sqrt{\pi} r_2+\sqrt{\pi} r_3$

$=\sqrt{\pi}\left(r_1+r_2+r_3\right)=$ RHS

Option (b), $LHS=\frac{1}{\sqrt{A_1}}+\frac{1}{\sqrt{A_2}}+\frac{1}{\sqrt{A_3}}=\frac{1}{\sqrt{\pi}}\left(\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}\right)$

$=\frac{1}{\sqrt{\pi}}\cdot\frac{1}{r}=\frac{1}{\sqrt{\pi r^2}} \quad \left(\because \frac{1}{r}=\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{I_3}\right)$

$=\frac{1}{\sqrt{A}}=$ RHS

Option (c), LHS $=\frac{1}{\sqrt{A_1}}+\frac{1}{\sqrt{A_2}}+\frac{1}{\sqrt{A_3}}=\frac{1}{\sqrt{\pi}\cdot r}$ [using option (b)]

RHS $=\frac{s^2}{\sqrt{\pi} r_1 r_2 r_3}=\frac{s^2}{\sqrt{\pi}\cdot\frac{\Delta}{s-a}\cdot\frac{\Delta}{s-b}\cdot\frac{\Delta}{s-c}}$

$=\frac{s^2(s-a)(s-b)(s-c)}{\sqrt{\pi}\Delta^3}=\frac{s\Delta^2}{\sqrt{\pi}\Delta^3}$

$= \frac{1}{\sqrt{\pi}} \cdot \frac{s}{\Delta} = \frac{1}{\sqrt{\pi}} \cdot \frac{1}{r} = LHS \quad \left(\because r = \frac{\Delta}{s}\right)$

Option (d) $LHS = \sqrt{A_1} + \sqrt{A_2} + \sqrt{A_3}$

$$= \sqrt{\pi}\,(r_1 + r_2 + r_3) \quad \text{[using option (a)]}$$
$$= \sqrt{\pi}(4R + r) \quad (\because 4R - r_1 + r_2 + r_3 - r)$$
$$= RHS$$

Q19 (2,4) - Multiple Correct

We know that,

$$r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b}, r_3 = \frac{\Delta}{s-c}$$
$$2r_2 = \frac{2\Delta}{s-b}, 3r_3 = \frac{3\Delta}{s-c}$$

We have,

$r_1 = 2r_2 = 3r_3 = k \quad \text{(let)}$

$\Rightarrow \frac{\Delta}{s-a} = \frac{2\Delta}{s-b} = \frac{3\Delta}{s-c} = k$

$\Rightarrow s - a = \frac{\Delta}{k}$ ...(i)

$s - b = \frac{2\Delta}{k}$ ...(ii)

$s - c = \frac{3\Delta}{k}$ ...(iii)

Now,

$$s - a + s - b + s - c = \frac{\Delta}{k} + \frac{2\Delta}{k} + \frac{3\Delta}{k}$$
$$3s - (a+b+c) = \frac{6\Delta}{k}$$
$$s = \frac{6\Delta}{k}$$

Putting the value of s in Eqs. (i), (ii) and (iii), we get

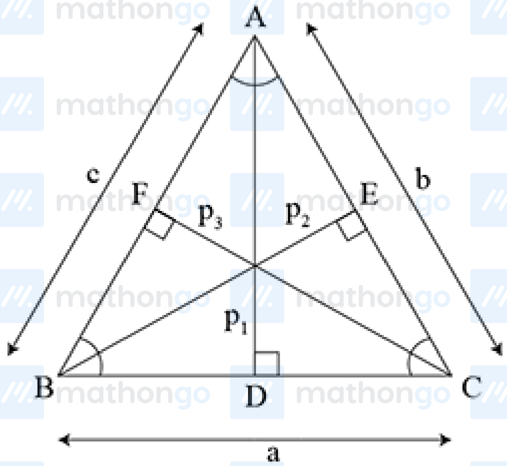
$$a = \frac{5\Delta}{k}, b = \frac{4\Delta}{k}, c = \frac{3\Delta}{k}$$
$$\therefore \quad a : b : c = 5 : 4 : 3$$

So, $a : b = 5 : 4$ and $a : c = 5 : 3$

Q20 (2,4) - Multiple Correct

$\because \Delta = \frac{1}{2}ap_1 = \frac{1}{2}bp_2 = \frac{1}{2}cp_3$ (by figure)

$\Rightarrow \quad p_1 = \frac{2\Delta}{a}, p_2 = \frac{2\Delta}{b}, p_3 = \frac{2\Delta}{c}$



But given that a, b, c are in AP.

Then, $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in HP.

$\Rightarrow \frac{2\Delta}{a}, \frac{2\Delta}{b}, \frac{2\Delta}{c}$ are also in HP.

$\Rightarrow p_1, p_2, p_3$ are in HP. [option (b)]

$\because \quad p_1 = \frac{2\Delta}{a}, \quad p_2 = \frac{2\Delta}{b}, \quad p_3 = \frac{2\Delta}{c}$

$\Rightarrow \quad p_1 = \frac{2\Delta}{2R\sin A}, \quad p_2 = \frac{2\Delta}{2R\sin B}, \quad p_3 = \frac{2\Delta}{2R\sin C}$

$\Rightarrow \quad \frac{1}{p_1} = \frac{R\sin A}{\Delta}, \quad \frac{1}{p_2} = \frac{R\sin B}{\Delta}, \quad \frac{1}{p_3} = \frac{R\sin C}{\Delta}$

$\because a, b, c$ are in AP.

Then $\sin A, \sin B, \sin C$ are also in AP.

$\Rightarrow \quad 2\sin B = \sin A + \sin C$ ...(i)

Now, $\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} = \frac{R}{\Delta}(\sin A + \sin B + \sin C)$

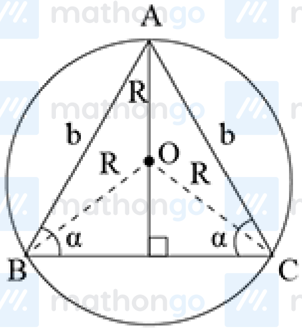
$= \frac{R}{\Delta} 3\sin B$ [from Eq.(i)]

$\leq \frac{3R}{\Delta}$

Q21 (1,3,4) - Multiple Correct

$\because \quad R = \frac{b}{2\sin\alpha} = \frac{b}{2}\operatorname{cosec}\alpha$ ...(i)

[option (a)]



$\because \angle B=\alpha, \angle C=\alpha,$ then $\angle A=180^{\circ}-2\alpha$

$\therefore \quad \Delta=\frac{1}{2} b \cdot b \sin \left(180^{\circ}-2 \alpha\right)=\frac{1}{2} b^2 \sin 2 \alpha$

and $r=4 R \sin \frac{\alpha}{2} \sin \frac{\alpha}{2} \sin \left(90^{\circ}-\alpha\right)$

$=4 \cdot \frac{b}{2 \sin \alpha} \cdot \sin ^2 \frac{\alpha}{2} \cdot \cos \alpha \cdot \frac{\sin \alpha}{\sin \alpha}$ [from Eq.(i)]

$=\frac{b \cdot 2 \sin ^2 \frac{\alpha}{2} \cdot 2 \sin \alpha \cdot \cos \alpha}{2 \cdot \sin ^2 \alpha}$

$=\frac{b(1-\cos \alpha) \sin 2 \alpha}{2\left(1-\cos ^2 \alpha\right)}$

$r=\frac{b \sin 2 \alpha}{2(1+\cos \alpha)}$ [(option (c)]

$\because \quad O I=\sqrt{R^2-2 R r}=R \sqrt{1-\frac{2 r}{R}}$

$=R \sqrt{1-4 \cos \alpha+4 \cos ^2 \alpha}$

$=R(2 \cos \alpha-1)$

$=R\left[2\left(2 \cos ^2 \frac{\alpha}{2}-1\right)-1\right]$

$=R\left[4 \cos ^2 \frac{\alpha}{2}-3\right]$

$=\frac{R\left[4 \cos ^3 \frac{\alpha}{2}-3 \cos \frac{\alpha}{2}\right]}{\cos \frac{\alpha}{2}}$

$=\frac{b}{2 \sin \alpha} \cdot \frac{\cos \frac{3 \alpha}{2}}{\cos \frac{\alpha}{2}}$ [from Eq.(i)]

$\therefore \quad O I=\left|\frac{b \cos \frac{3 \alpha}{2}}{2 \sin \alpha \cos \frac{\alpha}{2}}\right|$ [option (d)]

Q22 (1,2) - Multiple Correct

Given, $a=\alpha-\beta, b=\alpha+\beta, c=\sqrt{3 \alpha^2+\beta^2}$

$\because \cos C=\frac{a^2+b^2-c^2}{2 a b}$

$=\frac{(\alpha-\beta)^2+(\alpha+\beta)^2-\left(\sqrt{3 \alpha^2+\beta^2}\right)^2}{2(\alpha-\beta)(\alpha+\beta)}$

$=\frac{\alpha^2+\beta^2-2 \alpha \beta+\alpha^2+\beta^2+2 \alpha \beta-3 \alpha^2-\beta^2}{2\left(\alpha^2-\beta^2\right)}$

$=\frac{-\left(\alpha^2-\beta^2\right)}{2\left(\alpha^2-\beta^2\right)}=-\frac{1}{2}=\cos \frac{2 \pi}{3}$

$\therefore \quad C=\frac{2\pi}{3}$

$\therefore \quad A+B=\pi-C=\pi-\frac{2\pi}{3}=\frac{\pi}{3}$

Q23 (1,2,3) - Multiple Correct

We have,

$\frac{a+b}{11}=\frac{b+c}{13}=\frac{c+a}{12}=\lambda$ (let)

Then, $a+b=11\lambda$ ...(i)

$b+c=13\lambda \quad$ ...(ii)

$c+a=12\lambda \quad$ ...(iii)

On adding Eqs. (i), (ii) and (iii), we get

$2(a+b+c)=36\lambda$

$a+b+c=18\lambda \quad$ ...(iv)

On solving Eqs. (i) and (iv), (ii) and (iv), (iii) and (iv), we get

$c=7\lambda, a=5\lambda, b=6\lambda$

Now, $\cos A=\frac{b^2+c^2-a^2}{2bc}$

$=\frac{36\lambda^2+49\lambda^2-25\lambda^2}{2\times6\lambda\times7\lambda}$

$=\frac{36+49-25}{2\times7\times6}=\frac{5}{7}$

$\therefore \quad \frac{\cos A}{\frac{5}{7}}=1$ ...(v)

Similarly, $\frac{\cos B}{19/35}=1$ and $\frac{\cos C}{1/5}=1$

$\therefore \frac{\cos A}{5/7}=\frac{\cos B}{19/35}=\frac{\cos C}{1/5}$

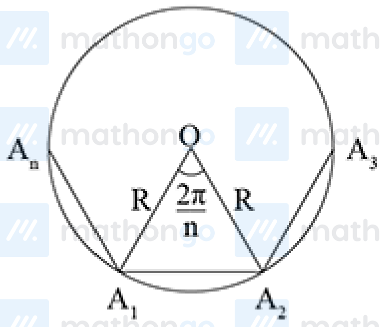
$\frac{\cos A}{25}=\frac{\cos B}{19}=\frac{\cos C}{7}$

But given that $\frac{\cos A}{p}=\frac{\cos B}{q}=\frac{\cos C}{r}$

$\therefore p=25, q=19, r=7$

Q24 (4) - Paragraph 1

From Fig. (i),



(i)

$$(A_1A_2)^2 = R^2 + R^2 - 2R \cdot R\cos\frac{2\pi}{n}$$
$$= 2R^2 - 2R^2\cos\frac{2\pi}{n}$$
$$= 2R^2\left(1 - \cos\frac{2\pi}{n}\right)$$
$$= 4R^2\sin^2\frac{\pi}{n}$$

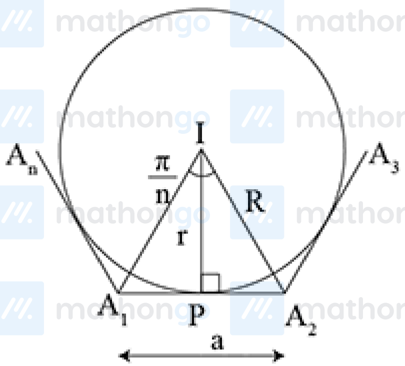
$A_1A_2 = 2R\sin\frac{\pi}{n}$

$\therefore a = 2R\sin\frac{\pi}{n}$

$\Rightarrow R = \frac{a}{2\sin\frac{\pi}{n}}$ ...(i)

From Fig. (ii),

From in ΔA_1PI, $\tan\frac{\pi}{n} = \frac{A_1P}{PI}$



(ii)

$$\tan\frac{\pi}{n} = \frac{\frac{a}{2}}{r}$$
$$r = \frac{1}{2}a\cot\frac{\pi}{n} \quad \text{...(ii)}$$

Now,

$$\begin{aligned} r+R &= \frac{1}{2} a \cot \frac{\pi}{n}+\frac{a}{2 \sin \frac{\pi}{n}} \\ &= \frac{a}{2} \cdot \frac{\cos \frac{\pi}{n}}{\sin \frac{\pi}{n}}+\frac{a}{2 \sin \frac{\pi}{n}} \\ &= \frac{a}{2 \sin \frac{\pi}{n}}\left(1+\cos \frac{\pi}{n}\right) \\ &= \frac{a}{2 \cdot 2 \sin \frac{\pi}{2n} \cos \frac{\pi}{2n}} \times 2 \cos^2 \frac{\pi}{2n} \\ &= \frac{a}{2} \cot \frac{\pi}{2n} \end{aligned}$$

Q25 (1) - Paragraph 1

Area of the regular polygon

$= n \times \operatorname{ar}(\Delta A_1 O A_2)$

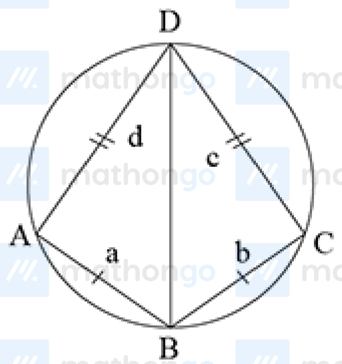
$= n \times \frac{1}{2} O A_1 \cdot O A_2 \cdot \sin \angle A_1 O A_2$ [from Fig.(i)]

$= n \times \frac{1}{2} R \cdot R \cdot \sin \frac{2\pi}{n} = \frac{n}{2} R^2 \sin\left(\frac{2\pi}{n}\right)$

Q26 (4) - Paragraph 2

We have,

$\angle A+\angle C=180^{\circ}$



and $\angle B+\angle D=180^{\circ}$

Let $AB=BC=6$ cm

and $CD=DA=8$ cm

Then, $\Delta ABD \cong \Delta BDC$

$\therefore \angle A=\angle C=\frac{180^{\circ}}{2}=90^{\circ}$

$\therefore$ In ΔABD, using Pythagoras theorem

$$\begin{aligned} BD &= \sqrt{AB^2 + AD^2} \\ &= \sqrt{6^2 + 8^2} = 10 \end{aligned}$$

$\therefore$ Radius of circumcircle $= \dfrac{BD}{2} = 5$ cm

Q27 (3) - Paragraph 2

Here, $AB = BC = 5$ cm, $CD = DA = 12$ cm

$\therefore \quad \angle A = \angle C = 90^\circ$, then

$\operatorname{ar}(ABCD) = \operatorname{ar}(\Delta ABD) + \operatorname{ar}(\Delta BCD)$

$= \frac{1}{2} \cdot AB \cdot AD + \frac{1}{2} \cdot BC \cdot CD$

$= \frac{1}{2} \times 5 \times 12 + \frac{1}{2} \times 5 \times 12$

$= 60 \text{ cm}^2$

$\therefore \quad AB + AD = BC + CD$, then incircle can be drawn

$s = \frac{a+b+c+d}{2} = \frac{5+5+12+12}{2} = 17$ cm

We know that,

$\operatorname{ar}(ABCD) = r \cdot s$

$\Rightarrow 60 = r \cdot 17$

$\therefore r = \frac{60}{17}$ cm

Q28 (3) - Paragraph 3

$\frac{bx}{c} + \frac{cy}{a} + \frac{az}{b} = b \sin B + c \sin C + a \sin A$

$= \frac{b^2+c^2+a^2}{2R}$

So, $k = 2R$

Q29 (3) - Paragraph 3

$\because \cot A+\cot B+\cot C$

$$=\frac{R}{abc}\left(b^2+c^2-a^2+c^2+a^2-b^2+a^2+b^2-c^2\right)$$
$$=\frac{R}{abc}\left(a^2+b^2+c^2\right)=\frac{R}{abc}\left(\frac{4\Delta^2}{x^2}+\frac{4\Delta^2}{y^2}+\frac{4\Delta^2}{z^2}\right)$$
$$=\frac{4R\Delta^2}{abc}\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)=\frac{4R\Delta}{abc}\cdot\Delta\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)$$
$$=\Delta\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)$$

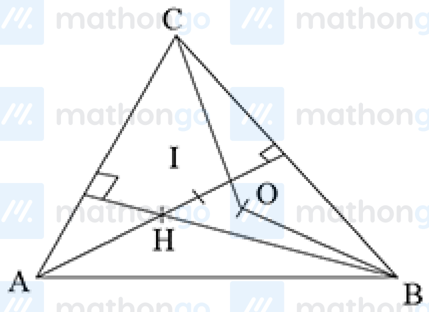
So, $k=\Delta$

Q30 (3) - Paragraph 4

Since, A, H, I, O and B are concyclic, so

$\angle AHB=\angle AIB=\angle AOB$

(angles of chord AB subtended at same side of AB in corresponding circle)



Here, $\angle AHB=\pi-\angle BAH-\angle HBA$

$$=\pi-\left(\frac{\pi}{2}-B\right)-\left(\frac{\pi}{2}-A\right)$$
$$=A+B=\pi-C$$

and $\angle AOB=2C$

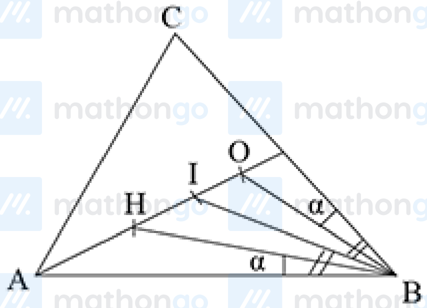
$\therefore \quad \pi-C=2C\Rightarrow C=\frac{\pi}{3}$

Q31 (2) - Paragraph 4

$\angle ABH=\frac{\pi}{2}-A$

and $\quad \angle OBC=\frac{\pi}{2}-A$

$\therefore \qquad \angle ABH=\angle OBC=\alpha$



Also, $\angle \mathrm{IBA}=\frac{B}{2}=\angle \mathrm{IBC}$

$\therefore \quad \mathrm{DIBH}=\angle IBO=\dfrac{B}{2}-\alpha$

$$=\frac{B}{2}-\left(\frac{\pi}{2}-A\right)$$

$\therefore$ HI and OI will be chords of equal lengths

Hence, $\frac{HI}{OI}=1$

Q32 (2) - Paragraph 5

Let a, b, c be the sides of a triangle such that

and $b+c=p$

$bc=q$

and $r=a$

We know that, AM $\geq$ GM

$$\frac{b+c}{2} \geq \sqrt{bc}$$

$$p \geq 2\sqrt{q}$$

$\therefore \angle A$ is the greatest angle.

It is given that, $(p+r)(p-r)=q$

$\Rightarrow (b+c+a)(b+c-a)=bc$

$$\Rightarrow 2s(2s-2a)=bc$$

$$\Rightarrow \frac{s(s-a)}{bc}=\frac{1}{4}$$

$$\Rightarrow \sqrt{\frac{s(s-a)}{bc}}=\frac{1}{2}$$

$$\Rightarrow \cos\frac{A}{2}=\cos 60^\circ$$

$$\Rightarrow \quad \frac{A}{2}=60^\circ$$

$$\therefore \angle A=120^\circ$$

Q33 (1) - Paragraph 5

We know that, area of $\Delta=\frac{1}{2}bc\sin A$

$=\frac{1}{2}q\sin 120^\circ \quad (\because bc=q)$

$=\frac{1}{2}q\cdot\frac{\sqrt{3}}{2}$

$=\frac{\sqrt{3}}{4}q$ sq units

Q34 (5) - Integer Type

Let $a_1 a+d_1 a+2d, a+3d$ be the sides of a quadrilateral

in AP.

Then, according to the question,

$a=6$

and $a+3d-a=6$ (difference)

$\Rightarrow d=2$

$\therefore$ Sides are $6, 8, 10, 12$.

Now, $2s=6+8+10+12$

$s=18$

Thus, $A^2=(18-6)(18-8)(18-10)(18-12)$

$=12\times 10\times 8\times 6$

$=5760$

We have,

$$A^2 = 1152\lambda$$
$$\Rightarrow 5760 = 1152\lambda$$
$$\Rightarrow \lambda = \frac{5760}{1152}$$
$$\therefore \lambda = 5$$

Q35 (2) - Integer Type

$\because \cot \frac{A}{2} \cdot \cot \frac{B}{2} = c$

$\sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \cdot \sqrt{\frac{s(s-b)}{(s-a)(s-c)}} = c$

$\frac{s}{s-c} = c$

$\Rightarrow \frac{1}{s-c} = \frac{c}{s} \quad \ldots(\text{i})$

Similarly, $\cot \frac{B}{2} \cot \frac{C}{2} = a$

$\Rightarrow \quad \frac{1}{s-a} = \frac{a}{s} \quad \ldots(\text{ii})$

and $\cot \frac{C}{2} \cot \frac{A}{2} = b$

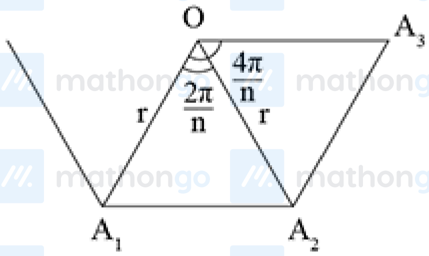
$\Rightarrow \frac{1}{s-b} = \frac{b}{s}$ ...(iii)

Now, on adding Eqs. (i), (ii) and (iii), we get

$$\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c} = \frac{a}{s} + \frac{b}{s} + \frac{c}{s}$$
$$= \frac{a+b+c}{s} = \frac{2s}{s} = 2$$

Q36 (7) - Integer Type

Let O and r be the centre and radius of circle passing through the vertices $A_1, A_2, A_3, \ldots, A_n$



$\because \quad OA_1 = OA_2 = r$

and $\cos \frac{2\pi}{n} = \frac{(OA_1)^2 + (OA_2)^2 - (A_1A_2)^2}{2OA_1 \cdot OA_2}$

$$\cos\frac{2\pi}{n}=\frac{2r^2-(A_1A_2)^2}{2r^2}$$
$$(A_1A_2)^2=2r^2-2r^2\cos\frac{2\pi}{n}$$
$$=2r^2\left(1-\cos\frac{2\pi}{n}\right)=2r^2\cdot 2\sin^2\frac{\pi}{n}$$

$\therefore\quad A_1A_2=2r\sin\frac{\pi}{n}$ ...(i)

Similarly, $A_1A_3=2r\sin\frac{2\pi}{n}$ ...(ii)

and $A_1A_4=2r\sin\frac{3\pi}{n}$ ...(iii)

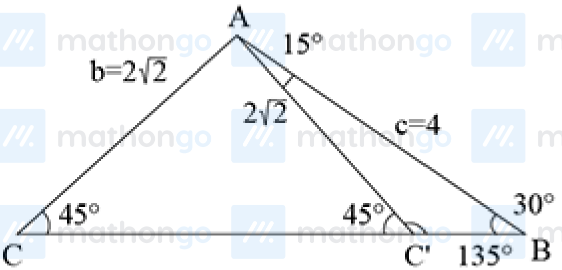
Now, we have,

$$\frac{1}{A_1A_2}=\frac{1}{A_1A_3}+\frac{1}{A_1A_4}$$
$$\Rightarrow\frac{1}{2r\sin\frac{\pi}{n}}=\frac{1}{2r\sin\frac{2\pi}{n}}+\frac{1}{2r\sin\frac{3\pi}{n}}$$
$$\Rightarrow\frac{1}{\sin\frac{\pi}{n}}=\frac{\sin\frac{3\pi}{n}+\sin\frac{2\pi}{n}}{\sin\frac{3\pi}{n}\sin\frac{2\pi}{n}}$$
$$\Rightarrow\sin\frac{3\pi}{n}\sin\frac{2\pi}{n}=\sin\frac{3\pi}{n}\sin\frac{\pi}{n}+\sin\frac{2\pi}{n}\sin\frac{\pi}{n}$$
$$\Rightarrow\quad\sin\frac{3\pi}{n}\sin\frac{2\pi}{n}-\sin\frac{2\pi}{n}\sin\frac{\pi}{n}=\sin\frac{3\pi}{n}\sin\frac{\pi}{n}$$
$$\Rightarrow\quad\sin\frac{2\pi}{n}\left(\sin\frac{3\pi}{n}-\sin\frac{\pi}{n}\right)=\sin\frac{3\pi}{n}\sin\frac{\pi}{n}$$
$$\Rightarrow\quad\left(2\sin\frac{2\pi}{n}\cos\frac{2\pi}{n}\right)\sin\frac{\pi}{n}=\sin\frac{3\pi}{n}\sin\frac{\pi}{n}$$
$$\Rightarrow\quad\sin\frac{4\pi}{n}=\sin\frac{3\pi}{n}\Rightarrow\sin\left(\frac{4\pi}{n}\right)=\sin\left(\pi-\frac{3\pi}{n}\right)$$
$$\Rightarrow\quad\frac{4\pi}{n}=\pi-\frac{3\pi}{n}$$
$$\Rightarrow\quad\frac{7}{n}=1\Rightarrow n=7$$

Q37 (4) - Integer Type

In ΔABC, using sine rule,

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$



$$\frac{a}{\sin A}=\frac{2\sqrt{2}}{\sin 30^\circ}=\frac{4}{\sin C}$$

If $\frac{2\sqrt{2}}{\sin 30^\circ}=\frac{4}{\sin C}$

$\Rightarrow \quad 4\sqrt{2}=\frac{4}{\sin C}\Rightarrow \sin C=\frac{1}{\sqrt{2}}=\sin 45^\circ$

$\Rightarrow \quad C=45^\circ$

$\therefore \quad A=180^\circ-(45^\circ+30^\circ)=105^\circ$

$\because \quad C=45^\circ,\ \text{then } \angle AC'C=45^\circ \quad (\because AC=AC')$

Now, $\angle AC'B=180^\circ-\angle AC'C$

$=180^\circ-45^\circ=135^\circ$

Now, $\operatorname{ar}(\Delta ABC)=\frac{1}{2}AB\times AC\cdot\sin A$

$=\frac{1}{2}\times 4\times 2\sqrt{2}\cdot\sin 105^\circ$

$=4\sqrt{2}\cdot\frac{\sqrt{3}+1}{2\sqrt{2}}=2(\sqrt{3}+1)$

and $\operatorname{ar}(\Delta ABC')=\frac{1}{2}AB\times AC'\cdot\sin 15^\circ$

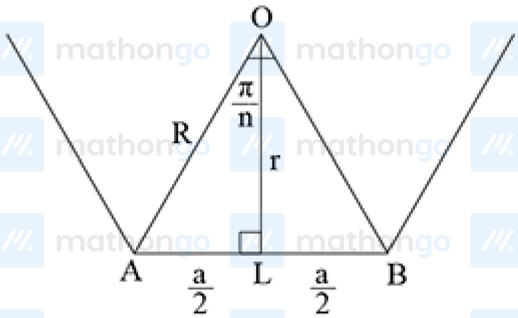
$=\frac{1}{2}\times 4\times 2\sqrt{2}\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)=2(\sqrt{3}-1)$

$\therefore$ Absolute difference of areas of triangle

$=|2\sqrt{3}+2-2\sqrt{3}+2|=4$sq unit

Q38 (5) - Integer Type

Let $OL=r$ and $OA=R, \angle AOL=\frac{\pi}{n}$ and a be the length of side of a regular polygon, then $AL=BL=\frac{a}{2}$



in ΔAOL

cosec $\frac{\pi}{n}=\frac{R}{\frac{a}{2}}\Rightarrow R=\frac{a}{2}\operatorname{cosec}\frac{\pi}{2}$ ...(i)

and cot $\frac{\pi}{2}=\frac{r}{\frac{a}{2}}\Rightarrow r=\frac{a}{2}\cot\frac{\pi}{n}$ ...(ii)

Now, $\quad \frac{R}{r}=\sqrt{5}-1 \quad$ (given)

$\Rightarrow \quad \frac{\frac{a}{2}\cdot\operatorname{cosec}\frac{\pi}{n}}{\frac{a}{2}\cdot\cot\frac{\pi}{n}}=\sqrt{5}-1$

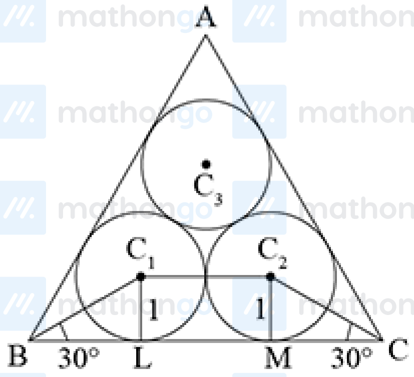
$\Rightarrow \quad \cos\frac{\pi}{n} = \frac{1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} = \frac{\sqrt{5}+1}{4}$

$\Rightarrow \quad \cos\frac{\pi}{n} = \cos\frac{\pi}{5} \quad \left(\because \cos\frac{\pi}{5} = \cos 36^\circ = \frac{\sqrt{5}+1}{4}\right)$

$\therefore \quad n = 5$

Q39 (3) - Integer Type

It is given that $C_1L = C_2M = 1$



In ΔBLC_1 and ΔCMC_2

$$\cot 30^\circ = \frac{BL}{C_1L} \text{ and } \cot 30^\circ = \frac{CM}{C_2M}$$

$$\sqrt{3} = \frac{BL}{1} \text{ and } \sqrt{3} = \frac{CM}{1}$$

$\therefore \quad BL = CM = \sqrt{3}$

and $LM = C_1C_2 = 1 + 1 = 2$

$\therefore BC = BL + LM + MC$

$= \sqrt{3} + 2 + \sqrt{3} = 2(1 + \sqrt{3})$

We know that area of equilateral Δ

$= \frac{\sqrt{3}}{4}(\text{Side})^2$

$= \frac{\sqrt{3}}{4}[2(1+\sqrt{3})]^2$

$= \frac{\sqrt{3}}{4} \cdot 4(1 + 3 + 2\sqrt{3})$

$= \sqrt{3}(4 + 2\sqrt{3})$

$= 6 + 4\sqrt{3}$ sq units

But $\Delta = 6 + 4\sqrt{p}$ (given)

$\therefore \quad p = 3$

Q40 (3) - Integer Type

We have, $r_1 = 2r_2 = 2r_3$

$$\frac{\Delta}{s-a} = \frac{2\Delta}{s-b} = \frac{3\Delta}{s-c} = \lambda \quad \text{(let)}$$

$\therefore \quad s-a = \frac{\Delta}{\lambda}, s-b = \frac{2\Delta}{\lambda}, s-c = \frac{3\Delta}{\lambda}$

Now, $s-a+s-b+s-c = \frac{\Delta}{\lambda} + \frac{2\Delta}{\lambda} + \frac{3\Delta}{\lambda}$

$$3s-(a+b+c) = \frac{6\Delta}{\lambda}$$

$$s = \frac{6\Delta}{\lambda} \quad (\because a+b+c = 2s)$$

$\therefore \quad s-a = \frac{\Delta}{\lambda}$

$a = s - \frac{\Delta}{\lambda} = \frac{6\Delta}{\lambda} - \frac{\Delta}{\lambda} = \frac{5\Delta}{\lambda}$

$b = s - \frac{2\Delta}{\lambda} = \frac{6\Delta}{\lambda} - \frac{2\Delta}{\lambda} = \frac{4\Delta}{\lambda}$

$c = s - \frac{3\Delta}{\lambda} = \frac{6\Delta}{\lambda} - \frac{3\Delta}{\lambda} = \frac{3\Delta}{\lambda}$

$\therefore \quad a:b:c = 5:4:3$

Thus, $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = \frac{5}{4} + \frac{4}{3} + \frac{3}{5}$

$= \frac{75+80+36}{60} = \frac{191}{60}$

But $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = \frac{191}{20\lambda}$ (given)

$$\therefore \lambda = 3$$